

RESEARCH REPORT

AI-Based Multi-Disease Risk Prediction Using Machine Learning

Abstract

Chronic diseases such as diabetes and cardiovascular diseases are increasing rapidly worldwide, especially in developing countries. Early detection plays a crucial role in reducing complications and improving patient outcomes. This project aims to develop a machine learning-based system to predict disease risk using simple clinical parameters such as age, body mass index (BMI), blood pressure, and glucose levels. Logistic Regression and Random Forest models were implemented using Orange Data Mining software. The models achieved an accuracy of approximately 60–70%. The results demonstrate that machine learning can be used as a supportive screening tool for early disease detection.

Introduction

Chronic diseases like diabetes and heart disease are among the leading causes of morbidity and mortality globally. In many healthcare settings, especially in low-resource environments, access to advanced diagnostic tools is limited. Therefore, there is a need for simple, cost-effective, and accessible methods for early disease prediction.

Artificial Intelligence (AI) and Machine Learning (ML) offer powerful tools to analyze patient data and identify patterns that can help predict disease risk. This project focuses on developing a simple AI-based system using basic clinical parameters.

Objective

To develop a machine learning model that predicts the risk of:

- Diabetes
- Heart disease

The system utilizes simple and easily available clinical data for these predictions.

Methodology

1. Data Collection

Two primary datasets were utilized for model training and testing:

- Diabetes dataset
- Heart disease dataset

2. Data Preprocessing

Relevant features were selected to ensure the model focuses on the most impactful clinical parameters:

- Age, BMI, and Blood Pressure
- Glucose and Insulin levels
- Number of Pregnancies (where applicable)

Unnecessary variables were removed, and target variables were converted into categorical form (Yes/No) for binary classification.

3. Model Development

Two distinct machine learning models were applied for comparative analysis:

- **Logistic Regression:** A statistical model used for binary classification.
- **Random Forest:** An ensemble learning method using multiple decision trees.

4. Model Evaluation

Performance was measured using Accuracy, Confusion Matrix analysis, and ROC Curve plotting.

Results

Model	Observed Accuracy
Logistic Regression	~60–67%
Random Forest	~62–65%

Confusion Matrix Analysis

- Successful identification of most non-disease cases.

- Reasonable detection of disease cases.
- Observations included some instances of false positives and false negatives.

ROC Curve

The ROC curve lies consistently above the diagonal line, indicating acceptable predictive performance for a screening-level tool.

Discussion

The results indicate that machine learning models can predict disease risk with moderate accuracy. Random Forest showed slightly better performance compared to Logistic Regression in certain test cases. The model performance can be significantly improved by using larger datasets and implementing more rigorous data preprocessing techniques.

Conclusion

This project demonstrates that machine learning can be effectively used to predict disease risk using simple clinical parameters. Although the accuracy is currently moderate, the system can serve as a useful supportive screening tool in clinical and low-resource settings where more complex diagnostics may be unavailable.

Future Improvements

- Integration of larger and more diverse datasets.
- Refinement of data cleaning and normalization techniques.
- Application of advanced machine learning models (e.g., Neural Networks).
- Development of a real-time application for direct clinical use.

FINAL STATEMENT: This model is intended for screening purposes only and not for medical diagnosis.